

SALESFORCE TRAILHEAD

Real-Time MNC Project Guide

IT Service Desk & Vendor Management System

Covers: Objects · Fields · Flows · Validation Rules · Profiles · Roles
Page Layouts · Approval Processes · Email-to-Case · Escalation Rules
Queue Membership · Reports & Dashboards · and more

SECTION 1 — PROJECT OVERVIEW

Project Name: IT Service Desk & Vendor Management System (ITSD-VMS)

This project simulates a real-world MNC (Multi-National Corporation) internal IT platform built entirely on Salesforce. The system manages IT service requests, vendor contracts, asset procurement, change requests, employee IT access provisioning, and SLA-driven case management — all within a single Salesforce org. Every Salesforce concept from core administration to advanced automation is covered.

Business Context

GlobalTech Corp (fictional MNC) has 5,000+ employees across 12 countries. The IT department handles 10,000+ service requests per month from employees, manages 200+ active vendors, maintains an IT asset inventory of 15,000+ devices, and processes 500+ change requests per quarter. The company needs a Salesforce-based system to centralize all these operations, enforce approval hierarchies, and meet SLA commitments.

Core Modules

#	Module	Purpose
1	IT Service Requests (Cases)	Employee-raised tickets via Email-to-Case or portal
2	Vendor Management	Manage vendors, contracts, and performance scores
3	IT Assets	Track hardware/software assigned to employees
4	Change Requests	IT change control with multi-level approvals
5	Procurement Orders	Purchase requests for new hardware/software
6	Employee IT Profile	IT-specific employee data, access levels, onboarding tasks
7	SLA & Escalation	Auto-escalation when SLAs are breached
8	Reports & Dashboards	KPIs for IT leadership

Key Salesforce Concepts Covered

- Custom Objects, Fields, Relationships (Lookup, Master-Detail, Many-to-Many)
- Profiles, Roles, Permission Sets, Field-Level Security
- Page Layouts, Record Types, Compact Layouts
- Validation Rules, Formula Fields, Roll-Up Summary Fields
- Flows (Screen Flow, Record-Triggered Flow, Scheduled Flow, Auto-launched Flow)
- Approval Processes with multi-step, dynamic approvers
- Queue Membership and Assignment Rules
- Email-to-Case, Entitlements, Escalation Rules, Macros
- Reports, Dashboards, Report Types
- Email Alerts, Tasks, Workflow (legacy awareness)
- Lightning App Builder, App Manager, Navigation

SECTION 2 — DATA MODEL (Objects & Relationships)

Standard Objects Used

- Case (IT Service Requests — core)
- Contact (Employee record)
- Account (Vendor record)
- User (Salesforce Users)

Custom Object 1: IT_Asset__c — IT Asset Registry

Field Label	API Name	Field Type	Purpose
Asset Name	Name	Auto Number (AST-{0000})	Unique asset ID
Asset Type	Asset_Type__c	Picklist	Laptop/Desktop/Mobile/Software/Peripheral
Serial Number	Serial_Number__c	Text(50), Unique	Hardware serial
Manufacturer	Manufacturer__c	Text(100)	Dell, HP, Apple etc.
Purchase Date	Purchase_Date__c	Date	When purchased
Warranty Expiry	Warranty_Expiry__c	Date	Warranty end date
Asset Status	Asset_Status__c	Picklist	Available/Assigned/In Repair/Retired
Assigned To	Assigned_To__c	Lookup → Contact	Employee assigned to
Vendor	Vendor__c	Lookup → Account	Vendor who supplied
Purchase Cost	Purchase_Cost__c	Currency	Cost in org currency
Depreciated Value	Depreciated_Value__c	Formula (Currency)	Auto-calculated depreciation
Location	Location__c	Text(100)	Office/Floor/Country
Is Active	Is_Active__c	Checkbox	Active flag
Procurement Order	Procurement_Order__c	Lookup → Procurement	Linked purchase order

Custom Object 2: Change_Request__c — IT Change Control

Field Label	API Name	Field Type	Purpose
Change Request Name	Name	Auto Number (CHG-{0000})	Unique CR ID
Title	Title__c	Text(150)	Short description
Description	Description__c	Long Text Area	Detailed change desc.
Change Type	Change_Type__c	Picklist	Standard/Emergency/Major
Priority	Priority__c	Picklist	Low/Medium/High/Critical
Status	Status__c	Picklist	Draft/Submitted/In Review/Approved/Rejected/Implemented
Risk Level	Risk_Level__c	Picklist	Low/Medium/High
Requested By	Requested_By__c	Lookup → User	Submitter
Assigned Technician	Assigned_Technician__c	Lookup → User	Tech owner
Planned Start	Planned_Start__c	DateTime	Implementation window start
Planned End	Planned_End__c	DateTime	Implementation window end

Field Label	API Name	Field Type	Purpose
Rollback Plan	Rollback_Plan__c	Long Text Area	Steps to undo if failed
Approval Status	Approval_Status__c	Picklist (read-only)	Managed by Approval Process
Related Asset	Related_Asset__c	Lookup → IT_Asset__c	Affected asset
Impact Assessment	Impact_Assessment__c	Rich Text Area	Business impact details

Custom Object 3: Vendor_Contract__c — Vendor Contracts

Field Label	API Name	Field Type	Purpose
Contract Name	Name	Auto Number (VCT-{0000})	Unique contract ID
Vendor	Vendor__c	Master-Detail → Account	Parent vendor (Account)
Contract Type	Contract_Type__c	Picklist	AMC/Support/SaaS/Hardware Supply
Start Date	Start_Date__c	Date	Contract begins
End Date	End_Date__c	Date	Contract expires
Contract Value	Contract_Value__c	Currency	Total value
Payment Terms	Payment_Terms__c	Picklist	Monthly/Quarterly/Annual
SLA Terms	SLA_Terms__c	Long Text Area	SLA clauses
Status	Status__c	Picklist	Draft/Active/Expired/Terminated
Renewal Reminder	Renewal_Reminder__c	Checkbox	Trigger 60-day alert
Account Manager	Account_Manager__c	Lookup → Contact	Vendor contact person
Performance Score	Performance_Score__c	Number(3,1)	Manually entered 1-10
Days to Expiry	Days_to_Expiry__c	Formula (Number)	TODAY() – End Date

Custom Object 4: Procurement_Order__c — Purchase Requests

Field Label	API Name	Field Type	Purpose
Order Name	Name	Auto Number (PRO-{0000})	Unique PO ID
Requested By	Requested_By__c	Lookup → User	Employee requesting
Department	Department__c	Picklist	IT/Finance/HR/Ops etc.
Item Description	Item_Description__c	Long Text Area	What is being requested
Quantity	Quantity__c	Number	Units needed
Estimated Cost	Estimated_Cost__c	Currency	Budget estimate
Approved Cost	Approved_Cost__c	Currency	Approved by Finance
Vendor	Vendor__c	Lookup → Account	Preferred vendor
Status	Status__c	Picklist	Draft/Submitted/Approved/Ordered/Received/Rejected
Urgency	Urgency__c	Picklist	Routine/Urgent/Emergency
Approval Status	Approval_Status__c	Picklist (read-only)	Managed by Approval Process
Budget Code	Budget_Code__c	Text(20)	Finance GL code
Received Date	Received_Date__c	Date	When PO received
Linked Assets	Linked_Assets__c	Roll-Up Count → Asset	Assets created from this PO

Custom Object 5: IT_Onboarding_Task__c — Onboarding Checklist

Field Label	API Name	Field Type	Purpose
Task Name	Name	Text(100)	Task label
Employee	Employee__c	Master-Detail → Contact	Parent employee
Task Type	Task_Type__c	Picklist	Account Setup/Hardware/Software/Access/Training/Badge
Status	Status__c	Picklist	Pending/In Progress/Completed/Blocked
Assigned To	Assigned_To__c	Lookup → User	IT staff responsible
Due Date	Due_Date__c	Date	Completion target
Completed Date	Completed_Date__c	Date	Actual completion
Notes	Notes__c	Long Text Area	Additional context
Is Mandatory	Is_Mandatory__c	Checkbox	Required before Day 1
Linked Asset	Linked_Asset__c	Lookup → IT_Asset__c	Asset assigned in task

Custom Object 6: Employee_IT_Profile__c — IT Access Record

Field Label	API Name	Field Type	Purpose
Profile Name	Name	Auto Number (EMP-{0000})	Unique ID
Employee	Employee__c	Lookup → Contact	Linked employee
Department	Department__c	Picklist	IT/HR/Finance/Ops etc.
Access Level	Access_Level__c	Picklist	Basic/Standard/Elevated/Admin
VPN Access	VPN_Access__c	Checkbox	VPN permission
Remote Work Enabled	Remote_Work_Enabled__c	Checkbox	WFH flag
IT Manager	IT_Manager__c	Lookup → User	Responsible IT manager
Onboarding Status	Onboarding_Status__c	Picklist	Not Started/In Progress/Completed
AD Account Created	AD_Account_Created__c	Checkbox	Active Directory
Email Provisioned	Email_Provisioned__c	Checkbox	Org email created
Software Licenses	Software_Licenses__c	Long Text Area	List of licenses
Total Assets	Total_Assets__c	Roll-Up Count → Asset	Count of assigned assets

Object Relationship Summary

Parent Object	Child Object	Relationship Type	Relationship Name
Account (Vendor)	Vendor_Contract__c	Master-Detail	Vendor Contracts
Account (Vendor)	IT_Asset__c	Lookup	Supplied Assets
Account (Vendor)	Procurement_Order__c	Lookup	Vendor Orders
Contact (Employee)	IT_Asset__c	Lookup	Assigned Assets
Contact (Employee)	IT_Onboarding_Task__c	Master-Detail	Onboarding Tasks
Contact (Employee)	Employee_IT_Profile__c	Lookup	IT Profile
Procurement_Order__c	IT_Asset__c	Lookup	Procured Assets
IT_Asset__c	Change_Request__c	Lookup	Related CRs
IT_Asset__c	IT_Onboarding_Task__c	Lookup	Task Asset
Case	IT_Asset__c	Lookup	Case Asset

SECTION 3 — PROFILES, ROLES & FIELD-LEVEL SECURITY

3.1 Roles Hierarchy

- CEO / CTO (top — visibility of all records)
- IT Director
- IT Manager (Change Requests, escalations)
- Sr. IT Engineer
- Jr. IT Engineer
- Vendor Manager
- Procurement Officer
- HR Manager
- HR Executive (onboarding tasks)
- Finance Manager
- Finance Analyst
- Employee (Standard User — self-service only)

3.2 Profiles to Create

Profile Name	Clone From	Purpose
IT_Admin_Profile	System Administrator	Full IT system control
IT_Manager_Profile	Standard User	Approve CRs, manage assets, view all cases
Sr_IT_Engineer_Profile	Standard User	Create/edit CRs, update assets, close cases
Jr_IT_Engineer_Profile	Standard User	View/update assigned cases and tasks only
Vendor_Manager_Profile	Standard User	Manage vendors, contracts, procurement
Procurement_Officer_Profile	Standard User	Create and submit procurement orders
HR_Manager_Profile	Standard User	View employee IT profiles, onboarding tasks
HR_Executive_Profile	Standard User	Create onboarding tasks, view assets
Finance_Manager_Profile	Standard User	Approve procurement costs, view contracts
Employee_Profile	Standard User	Raise cases, view own assets only

3.3 Field-Level Security Matrix

Legend: E = Edit | R = Read-Only | H = Hidden

Change_Request__c — Field Visibility by Profile

Field	IT Admin	IT Mgr	Sr Engr	Jr Engr	Employee	Finance Mgr
Title	E	E	E	R	R	R
Description	E	E	E	E	R	R
Change Type	E	E	E	R	H	R
Priority	E	E	E	R	H	H
Risk Level	E	E	R	R	H	R
Status	E	E	E	R	R	R
Rollback Plan	E	E	E	R	H	H
Approval Status	R	R	R	R	H	R
Assigned Technician	E	E	E	H	H	H
Impact Assessment	E	E	E	R	H	E

Procurement_Order__c — Field Visibility by Profile

Field	IT Admin	IT Mgr	Vendor Mgr	Proc. Officer	Finance Mgr	Employee
Item Description	E	E	R	E	R	E
Estimated Cost	E	E	R	E	R	R
Approved Cost	E	R	R	H	E	H
Vendor	E	E	E	E	R	H
Status	E	E	E	R	R	R
Urgency	E	E	R	E	R	E
Approval Status	R	R	R	R	R	H
Budget Code	E	R	R	H	E	H

IT_Asset__c — Field Visibility by Profile

Field	IT Admin	IT Mgr	Sr Engr	Jr Engr	HR Exec	Employee
Asset Type	E	E	E	R	R	R
Serial Number	E	E	E	R	H	H
Purchase Cost	E	R	H	H	H	H
Depreciated Value	E	R	H	H	H	H
Asset Status	E	E	E	E	R	R
Assigned To	E	E	E	R	R	R
Warranty Expiry	E	E	R	R	H	H
Vendor	E	R	R	H	H	H

3.4 Queue Membership

Queue Name	Object	Members (Profiles/Users)	Purpose
IT_L1_Support_Queue	Case	Jr. IT Engineers	First-line case triage
IT_L2_Support_Queue	Case	Sr. IT Engineers	Complex technical issues
IT_L3_Support_Queue	Case	IT Manager + Sr. Engineers	Escalated / VIP cases
Change_Review_Queue	Change_Request__c	IT Manager + IT Director	CR review board
Procurement_Queue	Procurement_Order__c	Procurement Officers	PO processing
Vendor_Contract_Queue	Vendor_Contract__c	Vendor Managers	Contract renewals
Onboarding_Queue	IT_Onboarding_Task__c	HR Executives + Jr. Engineers	New joiner tasks

SECTION 4 — PAGE LAYOUTS, RECORD TYPES & COMPACT LAYOUTS

4.1 Record Types

Object	Record Type Name	Profiles Assigned	Description
Case	IT_Hardware_Issue	All IT + Employee	Hardware breakdown tickets
Case	IT_Software_Issue	All IT + Employee	Software/app issues
Case	IT_Access_Request	All IT + Employee	Access provisioning
Case	IT_General_Enquiry	All profiles	Info requests
Change_Request__c	Standard_Change	IT Admin, Mgr, Sr Engr	Low-risk, pre-approved
Change_Request__c	Emergency_Change	IT Admin, IT Manager	Critical, fast-tracked
Change_Request__c	Major_Change	IT Admin, IT Manager	High-risk, full CAB review
Procurement_Order__c	Hardware_Purchase	Proc. Officer, IT Admin	New hardware PO
Procurement_Order__c	Software_License	Proc. Officer, IT Admin	Software licensing PO
Procurement_Order__c	Service_Contract	Vendor Mgr, IT Admin	New vendor contract PO

4.2 Page Layout Assignments

Each Record Type gets its own Page Layout, then assigned to profiles as follows:

Object / Layout Name	Profiles	Key Sections on Layout
Case — IT Hardware Layout	All IT, Employee	Case Info Asset Details SLA Timer Related Cases
Case — IT Software Layout	All IT, Employee	Case Info Software Details KB Articles Chatter
Case — Access Request Layout	IT Admin, HR Mgr, Employee	Case Info Access Fields Approval Section Tasks
CR — Standard Change Layout	IT Admin, Mgr, Sr Engr	CR Details Risk Rollback Related Assets Approvals
CR — Emergency Change Layout	IT Admin, IT Mgr	CR Details Urgency Approval Status Impact
PO — Hardware Layout	Proc. Officer, IT Admin	PO Info Vendor Asset Details Finance Approval
PO — Software Layout	Proc. Officer, IT Admin	PO Info License Count Vendor Cost Breakdown
Asset — Full Layout	IT Admin, IT Mgr, Sr Engr	All fields visible Related CRs Cases History
Asset — View Layout	Jr Engr, HR Exec, Employee	Limited fields Status Assigned To Related Cases

SECTION 5 — VALIDATION RULES

Object	Rule Name	Formula (Condition)	Error Message
Change_Request__c	CR_Rollback_Required_for_High_Risk	AND(Risk_Level__c = 'High', ISBLANK(Rollback_Plan__c))	Rollback Plan is mandatory for High Risk changes.
Change_Request__c	CR_Emergency_Needs_IT_Manager	AND(Change_Type__c = 'Emergency', NOT(\$Profile.Name = 'IT_Admin_Profile'), NOT(\$Profile.Name = 'IT_Manager_Profile'))	Only IT Admin or IT Manager can create Emergency Change Requests.
Change_Request__c	CR_Planned_End_After_Start	Planned_End__c <= Planned_Start__c	Planned End must be after Planned Start date.
Procurement_Order__c	PO_Cost_Limit_Employee	AND(\$Profile.Name = 'Employee_Profile', Estimated_Cost__c > 50000)	Employees cannot raise POs exceeding \$50,000. Contact IT Manager.
Procurement_Order__c	PO_Vendor_Required_on_Submit	AND(Status__c = 'Submitted', ISBLANK(Vendor__c))	A Vendor must be selected before submitting the Procurement Order.
IT_Asset__c	Asset_Retired_Cannot_Assign	AND(Asset_Status__c = 'Retired', NOT(ISBLANK(Assigned_To__c)))	Retired assets cannot be assigned to an employee.
IT_Asset__c	Asset_Serial_Length	AND(NOT(ISBLANK(Serial_Number__c)), LEN(Serial_Number__c) < 6)	Serial Number must be at least 6 characters.
Vendor_Contract__c	Contract_End_After_Start	End_Date__c <= Start_Date__c	Contract End Date must be after Start Date.
Case	Case_Close_Needs_Resolution	AND(Status = 'Closed', ISBLANK(Description))	A Resolution description is required before closing a Case.
IT_Onboarding_Task__c	Task_Complete_Date_Required	AND(Status__c = 'Completed', ISBLANK(Completed_Date__c))	Completed Date must be filled when marking a task as Completed.

SECTION 6 — FLOWS (All Types)

6.1 Record-Triggered Flows

Flow Name	Object & Trigger	Logic	Actions
Auto_Create_Onboarding_Tasks	Employee_IT_Profile__c After Insert / After Update When Onboarding_Status__c changes to 'In Progress'	Check: Onboarding_Status__c changed to 'In Progress'	Create 6 IT_Onboarding_Task__c records (Account Setup, Laptop Provisioning, Email Setup, VPN Access, Software Install, Badge & Access Card) assigned to Onboarding_Queue
Auto_Create_Asset_From_PO	Procurement_Order__c After Update When Status__c changes to 'Received'	Loop Quantity__c times	Create IT_Asset__c record per unit with Asset_Status__c = 'Available', linked to PO and Vendor
Send_CR_Approval_Email	Change_Request__c After Update When Approval_Status__c changes	Decision: Approved vs Rejected	Email Alert to Requested_By__c with outcome + IT Director CC if Rejected
Update_Asset_Status_on_Case	Case After Update When Status = 'In Progress' and Related Asset is not blank	Get Related Asset	Update Asset_Status__c to 'In Repair'
Retire_Asset_Cleanup	IT_Asset__c Before Update When Asset_Status__c changes to 'Retired'	—	Null out Assigned_To__c, trigger notification to HR
Contract_Expiry_Flag	Vendor_Contract__c After Update	If Days_to_Expiry__c <= 60 AND Status__c = 'Active'	Set Renewal_Reminder__c = true, create Task for Vendor Manager

6.2 Scheduled Flows

Flow Name	Schedule	Logic	Action
SLA_Breach_Alert_Flow	Every hour	Query Cases: Status != Closed AND CreatedDate <= NOW() - 24hrs AND Owner != Queue	Send Email Alert to Case Owner + IT Manager; Update Case Priority to 'Critical'
Contract_Renewal_Reminder_Flow	Daily at 8 AM	Query Vendor_Contract__c where Days_to_Expiry <= 60 and Renewal_Reminder = true	Create Task for Vendor Manager; Send Email to IT Director
Asset_Warranty_Alert_Flow	Weekly Monday	Query IT_Asset__c where Warranty_Expiry <= TODAY + 30	Create Task for IT Manager with asset details
Pending_PO_Reminder_Flow	Daily	Query Procurement_Order__c where Status = Submitted and CreatedDate < NOW() - 48hrs	Send reminder email to Finance Manager and Procurement Officer

6.3 Screen Flows (Self-Service)

Flow Name	Access Point	Screens	Outcome
Employee_Asset_Request_Flow	Employee Home Page button	1. Request Type 2. Asset Details 3. Justification 4. Confirm	Creates Procurement_Order__c with Status = Draft, assigned to Procurement Queue
IT_Onboarding_Wizard_Flow	HR Executive — Employee record	1. Employee Info 2. Access Level 3. Software 4. Hardware 5. Review	Creates Employee_IT_Profile__c + triggers Auto_Create_Onboarding_Tasks

Flow Name	Access Point	Screens	Outcome
Change_Request_Submission_Flow	IT Engineer — App Home	1. CR Type 2. Details 3. Risk 4. Rollback 5. Review	Creates Change_Request__c with Status = Draft; submits for approval if risk = High
Vendor_Performance_Review_Flow	Vendor Manager — Account page	1. Vendor 2. Contract 3. Score 4. Comments	Updates Performance_Score__c on Vendor_Contract__c and logs to Chatter

SECTION 7 — APPROVAL PROCESSES

7.1 Change Request Approval — Major Change (3 Steps)

Step	Approver	Entry Criteria	On Approve	On Reject
Step 1 — IT Manager Review	IT Manager (role-based)	Change_Type__c = 'Major'	Status → 'In Review' Send email to approver	Status → 'Rejected' Email to Requested_By
Step 2 — IT Director Approval	IT Director (role-based)	Approved at Step 1	Status → 'Approved' Send email to requester	Status → 'Rejected' Email chain to all
Step 3 — Finance Sign-off	Finance Manager (if cost > ■1L)	Approved at Step 2 AND Estimated_Cost > 1L	Final Approval_Status = 'Approved' Unlock record	Status → 'Rejected' Email Finance note

Initial Submission Actions: Lock record for editing | Set Approval_Status__c = 'Pending Approval' | Email IT Manager queue

Final Rejection: Unlock record | Set Status__c = 'Rejected' | Log rejection reason via Chatter Post action

7.2 Procurement Order Approval — 2 Steps

Step	Approver	Criteria	On Approve	On Reject
Step 1 — IT Manager	IT Manager (role-based)	Status__c = 'Submitted'	Approval_Status = 'Step 1 Approved' Send email	Status → 'Rejected' Email requester
Step 2 — Finance Mgr	Finance Manager (if > ■25K)	Step 1 approved AND Cost > ■25,000	Status → 'Approved' Unlock record Email Proc. Officer	Status → 'Rejected' Email chain

7.3 Emergency Change Request Approval — 1 Step (Fast Track)

Step	Approver	Entry Criteria	On Approve	On Reject
Step 1 — IT Director (Direct)	IT Director OR IT Admin	Change_Type__c = 'Emergency'	Status → 'Approved' Implementation can begin immediately Chatter post auto-created	Status → 'Rejected' Email requester + IT Manager

7.4 Approval Process — Recall & Delegation

- Recall: Submitter can recall a pending approval if Status is still 'Submitted' (enable 'Allow submitter to recall')
- Delegation: All approvers must set up delegate approvers in their user settings to handle leave scenarios
- Reminder Email: Set approval step reminder after 24 hours for POs; after 48 hours for CRs
- Chatter Approval: Enable 'Approve/Reject from Chatter' for mobile convenience

SECTION 8 — EMAIL-TO-CASE, SLA & ESCALATION

8.1 Email-to-Case Configuration

Setting	Value / Configuration
Feature	Email-to-Case (Setup → Email-to-Case)
Routing Email Address	itsupport@globaltech.com (create in org as routing address)
Case Origin	Auto-set to 'Email'
Default Case Owner	IT_L1_Support_Queue
Case Priority Default	Medium (can be overridden by Assignment Rule)
Email Subject → Case Subject	Mapped automatically
Email Body → Case Description	Mapped automatically
Sender Email → Contact Email	Match/create Contact automatically
Auto-Response Rule	Send acknowledgement email with Case Number on creation
Thread ID	Enable email threading to link replies to same Case

8.2 Case Assignment Rules

Rule Entry #	Criteria	Assign To
1	Subject contains 'VPN' OR 'network' OR 'firewall'	IT_L2_Support_Queue
2	Subject contains 'password' OR 'reset' OR 'access'	IT_L1_Support_Queue
3	Subject contains 'server' OR 'database' OR 'crash'	IT_L3_Support_Queue
4	Contact.Department = 'C-Suite' OR Contact.Title contains 'Director'	IT_L3_Support_Queue (VIP)
5	Origin = 'Email' (default catch-all)	IT_L1_Support_Queue

8.3 Entitlements & SLA Milestones

Create Entitlement Process: 'Standard IT SLA'

Milestone Name	Time Trigger	On Success Action	On Violation Action
First Response	Within 1 hour of case creation	Update milestone status	Email Alert to Case Owner + L1 Queue
Acknowledgement	Within 2 hours	Set Case status to 'In Progress'	Escalate to L2 Queue
Resolution Target	Within 8 hours (business hours)	Close milestone	Email IT Manager + elevate Priority to High
SLA Breach (Final)	Within 24 hours	—	Email IT Director + create Task on IT Manager

8.4 Escalation Rules (24-Hour Auto-Escalation)

Rule Name	Criteria	Hours	Action
L1_24hr_Escalation	Case Status != Closed AND Case Origin = Email AND Age > 24hrs	24 (business hrs)	Reassign to IT_L2_Support_Queue + Email IT Manager + Chatter Post

Rule Name	Criteria	Hours	Action
L2_48hr_Escalation	Case Status != Closed AND Owner = L2 Queue AND Age > 48hrs	48 (business hrs)	Reassign to IT_L3_Support_Queue + Email IT Director + Priority = Critical
VIP_4hr_Escalation	Contact.Title contains 'Director' AND Age > 4hrs	4 (business hrs)	Email IT Manager immediately + Priority = Critical

The 24-hour no-reply email trigger is handled by: (1) Escalation Rules above reassigning the case AND (2) a Scheduled Flow 'SLA_Breach_Alert_Flow' (Section 6.2) sending a targeted email to the Case Owner and CC-ing the IT Manager. Use Email Alerts with dynamic templates referencing the Case fields.

8.5 Auto-Response & Email Alert Templates

Template Name	Trigger	To	Subject	Key Body Fields
Case_Auto_ACK	Case Created (Email origin)	Contact (sender)	Re: Your IT Request {{CaseNumber}}	Case Number, Expected Response Time, L1 Queue Contact
SLA_Breach_24hr_Alert	Escalation Rule trigger	Case Owner + IT Mgr	ALERT: Case {CaseNumber} SLA Breached	Case Subject, Age, Priority, Employee Name
Case_Resolved_Notify	Case Status = Closed	Contact	Resolved: {Subject} {{CaseNumber}}	Resolution Summary, Satisfaction Survey Link
CR_Approval_Request	CR Approval Step 1	IT Manager	Action Required: CR {Name} Pending	CR Title, Risk, Planned Date, Rollback Plan
CR_Approved_Notify	CR Final Approval	Requested By	Approved: Change Request {Name}	Approval chain, Implementation Date
CR_Rejected_Notify	Any CR Rejection	Requested By + CC Mgr	Rejected: CR {Name} — Action Required	Rejection reason, Next steps
PO_Approved_Notify	PO Final Approval	Proc. Officer	PO {Name} Approved — Proceed to Order	Approved Cost, Vendor, Budget Code
Contract_Expiry_60day	Scheduled Flow	Vendor Manager	Contract Expiring in 60 Days: {Name}	Contract Value, SLA Terms, Vendor Contact

SECTION 9 — REPORTS & DASHBOARDS

9.1 Custom Report Types

Report Type Name	Primary Object	Related Object	Purpose
Cases with Assets	Case	IT_Asset__c	Cases showing related asset details
Employees with Onboarding Tasks	Contact	IT_Onboarding_Task__c	Onboarding completion tracking
Vendors with Contracts	Account	Vendor_Contract__c	Vendor portfolio overview
POs with Asset Records	Procurement_Order__c	IT_Asset__c	PO to asset fulfillment
CRs with Assets	Change_Request__c	IT_Asset__c	Change impact on assets

9.2 Key Reports

Report Name	Type	Grouped By	Metrics	Audience
Open Cases by Queue	Summary	Owner (Queue)	Count, Avg Age, % SLA Met	IT Manager
Cases Breaching SLA (24hr)	Tabular	—	Case#, Age, Owner, Priority	IT Director
CR Status Tracker	Summary	Status, Change Type	Count, Risk Distribution	IT Manager
Vendor Contract Expiry Dashboard	Summary	Vendor, End Date	Days to Expiry, Value	Vendor Manager
Asset Utilisation Report	Summary	Asset Type, Status	Count by Status	IT Admin
Procurement PO Pipeline	Summary	Status, Department	Total Cost, Pending Count	Finance Manager
Onboarding Task Completion	Matrix	Employee, Task Type	% Complete, Overdue Count	HR Manager
Monthly IT Incident Trend	Summary	Month (Created Date)	Case Count, Avg Resolution	IT Director

9.3 Dashboards

Dashboard Name	Audience	Components (widgets)
IT Operations Command Centre	IT Director + IT Manager	Open Cases Gauge SLA Breach Trend Line Cases by Queue Bar CR Status Donut Asset Status Pie
Vendor & Contract Tracker	Vendor Manager	Contracts Expiring (Table) Active Vendors Count Avg Performance Score Gauge PO Pipeline Funnel
HR Onboarding Dashboard	HR Manager	Onboarding Status Bar Tasks Overdue Count New Joiners this Month Asset Assignment Rate
Finance Procurement View	Finance Manager	Pending Approvals Count Total PO Value by Month Approved vs Rejected POs Budget Utilisation Gauge

SECTION 10 — LIGHTNING APP, NAVIGATION & HOME PAGES

10.1 Lightning App: 'IT Service Desk'

Setting	Value
App Name	IT Service Desk
App Label	IT Service Desk
App Type	Lightning App
Navigation Items	Home Cases IT Assets Change Requests Procurement Orders Vendor Contracts Onboarding Tasks Reports Dashboards
Assign to Profiles	IT_Admin, IT_Manager, Sr_IT_Engineer, Jr_IT_Engineer
Utility Bar	Recent Cases My Open Tasks Macros (for IT staff)

10.2 Lightning App: 'HR IT Onboarding'

Setting	Value
App Name	HR IT Onboarding
Navigation Items	Home Onboarding Tasks Employee IT Profiles Contacts Reports
Assign to Profiles	HR_Manager_Profile, HR_Executive_Profile

10.3 Lightning App: 'Vendor & Finance'

Setting	Value
App Name	Vendor & Finance
Navigation Items	Home Vendor Contracts Procurement Orders Accounts Reports Dashboards
Assign to Profiles	Vendor_Manager_Profile, Procurement_Officer_Profile, Finance_Manager_Profile

10.4 Lightning App: 'Employee Self-Service'

Setting	Value
App Name	Employee Self-Service
Navigation Items	Home My Cases My Assets New Request (Screen Flow)
Assign to Profiles	Employee_Profile
Home Page	Custom home page with: My Open Cases list Asset Request button IT Announcements component

SECTION 11 — FORMULA FIELDS & ROLL-UP SUMMARY FIELDS

11.1 Formula Fields

Object	Field Label	Return Type	Formula
IT_Asset__c	Depreciated_Value__c	Currency	Purchase_Cost__c * (1 - (YEAR(TODAY()) - YEAR(Purchase_Date__c)) * 0.2)
IT_Asset__c	Is_Under_Warranty__c	Checkbox	Warranty_Expiry__c >= TODAY()
Vendor_Contract__c	Days_to_Expiry__c	Number	End_Date__c - TODAY()
Vendor_Contract__c	Is_Expiring_Soon__c	Checkbox	Days_to_Expiry__c <= 60 && Days_to_Expiry__c > 0
Change_Request__c	Duration_Hours__c	Number	(Planned_End__c - Planned_Start__c) * 24
IT_Onboarding_Task__c	Days_Overdue__c	Number	IF(AND(Status__c != 'Completed', Due_Date__c < TODAY()), TODAY() - Due_Date__c, 0)
Procurement_Order__c	Cost_Variance__c	Currency	Approved_Cost__c - Estimated_Cost__c

11.2 Roll-Up Summary Fields

Parent Object	Field Label	Child Object	Roll-Up Type	Filter Criteria
Account (Vendor)	Total_Contract_Value__c	Vendor_Contract__c	SUM of Contract_Value__c	Status = 'Active'
Account (Vendor)	Active_Contract_Count__c	Vendor_Contract__c	COUNT	Status = 'Active'
Contact (Employee)	Total_Assets_Assigned__c	IT_Asset__c	COUNT	Asset_Status__c = 'Assigned'
Contact (Employee)	Pending_Tasks_Count__c	IT_Onboarding_Task__c	COUNT	Status__c != 'Completed'
Procurement_Order__c	Total_Assets_Created__c	IT_Asset__c	COUNT	All records
Employee_IT_Profile__c	Completed_Tasks__c	IT_Onboarding_Task__c	COUNT	Status__c = 'Completed'

SECTION 12 — RECOMMENDED BUILD SEQUENCE

Follow this exact sequence to avoid dependency errors:

Phase 1: Foundation

- Create all 6 Custom Objects (no fields yet) — just the object shell
- Enable Email-to-Case and configure routing address
- Create all Roles in the Hierarchy
- Create all Profiles (clone from Standard User / Sys Admin)
- Create Users and assign Profiles and Roles

Phase 2: Data Model

- Add all fields to each Custom Object (text, picklist, date, currency etc.)
- Add Lookup and Master-Detail relationship fields (child side references parent)
- Add Formula Fields (all dependencies on basic fields must exist first)
- Add Roll-Up Summary Fields (child objects and fields must exist)
- Create Custom Report Types

Phase 3: Security

- Set Object-Level Security (CRUD) on all profiles for all objects
- Set Field-Level Security per the matrix in Section 3.3
- Create Queues and add members
- Configure OWD (Org-Wide Defaults): Cases = Private, Assets = Private, CRs = Private, Vendor Contracts = Private
- Create Sharing Rules where role hierarchy alone is insufficient

Phase 4: Automation & Logic

- Create Validation Rules (objects must have fields first)
- Create Email Templates (needed by flows and approval processes)
- Create Email Alerts referencing the templates
- Build Record-Triggered Flows in order (Auto_Create_Onboarding_Tasks first, then others)
- Build Scheduled Flows
- Build Screen Flows
- Create Case Assignment Rules
- Create Case Auto-Response Rules
- Create Entitlement Process and Milestones
- Create Escalation Rules

Phase 5: Approvals & UI

- Build Approval Processes (email alerts and field updates must exist first)
- Create Record Types for Case, Change Request, Procurement Order
- Create Page Layouts per Record Type
- Assign Page Layouts to Profile + Record Type combinations
- Create Compact Layouts for each object
- Build Lightning Apps (IT Service Desk, HR IT Onboarding, Vendor & Finance, Employee Self-Service)
- Build Home Page Lightning Pages and assign to profiles/apps

Phase 6: Reports & Testing

- Create all Reports in dedicated report folders per audience
- Create Dashboards and set running user + refresh schedule
- Test Email-to-Case end-to-end: send email → case created → SLA timer → escalation
- Test Approval Processes: submit → approve Step 1 → approve Step 2 → verify field updates
- Test Flows: create Employee IT Profile → verify onboarding tasks created → verify asset created on PO receive
- Verify field-level security by logging in as each profile
- Test Validation Rules: try to violate each rule and confirm error message

SECTION 13 — TRAILHEAD ORG TIPS & LIMITATIONS

Trailhead Limitation	Workaround / Note
Max 2 active Flows in free org	Activate flows one at a time while testing; deactivate others temporarily
Email sending limited	Use Log a Case and test email templates via preview; use Email Simulator in Setup
No real org email domain	Use Salesforce's generated routing email for Email-to-Case testing
Approval email delivery may vary	Verify approval via Setup > Process Automation > Approval Processes > Pending Approvals
Scheduled Flows may not fire on free org	Use Debug button in Flow Builder to simulate scheduled trigger manually
Max 10 custom objects (Developer Org)	All 6 objects in this project fit within the limit
Max 3 custom profiles (some orgs)	Use Permission Sets for additional profile-level customization where needed
Reports limited without Einstein	All standard reports and dashboards in this guide work without Einstein Analytics

Recommended Trailhead Modules to Complete Alongside This Project

- Admin Beginner Trail — covers Setup, Users, Data Model
- Data Modeling → Trailhead module on Objects, Fields, Relationships
- Salesforce Flow → Automate business processes with Flow Builder
- Service Cloud Basics → Email-to-Case, Queues, Entitlements
- Approval Processes → Automate approvals with Process Builder and Flow
- Reports & Dashboards for Lightning Experience
- Security Basics → Profiles, Roles, Permission Sets, OWD, Sharing Rules
- Lightning App Builder → Build and customize Lightning pages

QUICK REFERENCE — OBJECT API NAMES & KEY PICKLIST VALUES

Object Label	API Name	Key Picklist Field	Values
IT Asset	IT_Asset__c	Asset_Status__c	Available Assigned In Repair Retired Lost
IT Asset	IT_Asset__c	Asset_Type__c	Laptop Desktop Mobile Server Printer Software Peripheral
Change Request	Change_Request__c	Change_Type__c	Standard Emergency Major
Change Request	Change_Request__c	Status__c	Draft Submitted In Review Approved Rejected Implemented Closed
Change Request	Change_Request__c	Risk_Level__c	Low Medium High Critical
Procurement Order	Procurement_Order__c	Status__c	Draft Submitted Step 1 Approved Approved Ordered Received Rejected
Procurement Order	Procurement_Order__c	Urgency__c	Routine Urgent Emergency
Vendor Contract	Vendor_Contract__c	Contract_Type__c	AMC SaaS Hardware Supply Support Consulting
Vendor Contract	Vendor_Contract__c	Status__c	Draft Active Under Renewal Expired Terminated
Onboarding Task	IT_Onboarding_Task__c	Task_Type__c	Account Setup Hardware Software VPN Access Training Badge & Access
Onboarding Task	IT_Onboarding_Task__c	Status__c	Pending In Progress Completed Blocked Cancelled
Employee IT Profile	Employee_IT_Profile__c	Access_Level__c	Basic Standard Elevated Admin Privileged
Employee IT Profile	Employee_IT_Profile__c	Onboarding_Status__c	Not Started In Progress Completed